

*Total No. of Questions : 8]*

*[Total No. of Printed Pages : 2*

**Roll No .....**

**EC-7005(2)-CBGS**

**B.E. VII Semester**

Examination, June 2020

**Choice Based Grading System (CBGS)**

**Wireless Communication**

***Time : Three Hours***

***Maximum Marks : 70***

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1.
  - a) Discuss briefly about the economic and social aspects to be considered while designing the system.
  - b) Discuss about the spectrum limitation and the basic principle of frequency reuse for unregulated access.
2. Discuss about noise limited systems and its link budget design.
3.
  - a) Derive the amplitude and phase distribution of the received signal for small scale fading with a dominant component.
  - b) Discuss about the temporal variations for moving mobile station.
4.
  - a) Discuss the characterization of deterministic linear time variant systems.
  - b) Discuss about ultra wideband signals with large relative Bandwidth.

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5.
  - a) Discuss about wideband models.
  - b) Explain the time domain measurement analysis.
  
6.
  - a) With the help of transceiver block diagram explain the wireless communication link.
  - b) Explain the concept of  $\pi/4$  differential quadrature phase shift keying.
  
7.
  - a) Explain about micro diversity, macro diversity and simulcast.
  - b) Discuss the principle of viterbi detector.
  
8. Write short notes on any two of the following:
  - a) Wave guiding
  - b) Deterministic channel modeling methods
  - c) Decision feedback equalizers

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